In the Name of God, the Most High

Faculty of Electrical and Computer Engineering

Isfahan University of Technology

Machine Learning - First Semester - 1401-02

Assignment No. 1 - Due Saturday, 1401/7/30

Introduction to Distributions and Python

Python is now considered one of the important programming languages in artificial intelligence. Therefore, try to become as familiar as possible with this language. This assignment aims to familiarize you further with Python and some distributions.

1. The scores of students in a course are provided in the file Data1mltak.xlsx (attached to this assignment in the system). The midterm and final exam scores are out of 10, while the other scores are out of 20. Plot a scatter matrix of these scores. In this plot, do not include the columns corresponding to the total score and course pass status. The scatter plot should show the points corresponding to each pair of scores. For an introduction to this plot, refer here. Try to color the points according to the value of the course pass column (pass).

Based on this plot, in your opinion, which pair of scores can better characterize whether a student passes or fails the course?

To read the information in this file, use the read_excel() function in the **pandas** package. To draw the scatter plot, you may also use the scatter_matrix() function in the same package. Also, study DataFrame in this package.

2. Fit a Gaussian distribution to the values in the total score column (total), obtain the parameters of that distribution, and plot the distribution using Python. To do this, become familiar with the matplotlib package. Use your own code to obtain the parameters.

3. Solve Question 1.30 from Bishop's book.

4. Derive the entropy of a Bernoulli distribution.

Submit your answers to the above questions and the results of your experiments as a typed report in PDF format, together with your programs, through the system. Also prepare and submit a short video with narration explaining the execution of your programs.

Please pay attention to the following points:

- Assignments may be submitted up to two days after the deadline; 10% of the grade will be deducted for each day of delay.

- Avoid colloquial and non-Persian writing, as it will result in a grade deduction. Do not include handwritten work or photographs of it in your report.

- Only assignments submitted through the system will be graded. Under no circumstances should you email your assignment.

- No credit will be given for assignments that are similar to, or downloaded from, other submissions.

Good luck
Palhang